



Group Work Form 2025

Please comment honestly and indicate if there are any concerns that you believe the teaching staff should consider in the assessment of your group project.

Team Member 1:	Chloe Fulbrook, 30210975	Email:	chloe.fulbrook@ucalgary.ca
Team member 2:	Devon Calvin, 30204532	Email:	devon.calvin@ucalgary.ca
Team member 3:	Ethan Lee, 30213225	Email:	ethan.lee4@ucalgary.ca
Team member 4:		Email:	

How do you rate your group's overall cohesion?

Poor		Average		Excellent
☐ 1	☐ 2	☐ 3	☐ 4	☒ 5

How would you distribute the overall contribution of your team members? (should sum to 100%)

TM1:	33 %	TM2:	33 %	TM3:	33 %	TM4:	%
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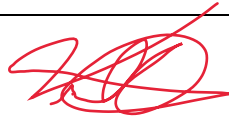
List the responsibilities/roles of each team member

TM1: Iterated and expanded the Lab 4 baseline code into the full project framework. Implemented the early integration of the FreeRTOS task structure and helped adapt the UART, ADC, and button modules into the project architecture. Assisted with debugging the initial system flow and validating that each module (UART, PWM LED, ADC sampling) functioned correctly before final FSM integration. Ensured the transition from the Lab 4 FSM to the multi-task FreeRTOS design maintained consistent behaviour and proper timing.	TM2: Verified FreeRTOS task interactions, including queue, semaphore, and interrupt behaviour. Debugged state-machine transitions for PB1/PB2/PB3 inputs, pause/resume logic, and end-state timing. Validated real-time output, LED behaviour, and display updates. Designed all major diagrams, including the main FreeRTOS FSM, the four pseudo-FSMs (LED2, button processing, ADC/PWM, and countdown), and the full FreeRTOS task-interaction diagram.
TM3: Performed verification of system behaviour across all modules (ADC, UART, LED PWM, button press detection, and timing). Ensured correctness of countdown logic, display formatting, and inter-task communication. Completed a major portion of the written report, including explanation of project module organization, requirement-D flow diagrams, and descriptions of task behaviour, ISR interactions, and timing considerations. Helped validate the full system against the functional requirements and project brief.	TM4:

General comments (you may attach additional sheets if required to provide more information)

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Signatures

TM1:  Date: 12/01/25	TM2:  Date: 12/01/25
TM3:  Date: 12/01/25	TM4: Date: